KIT 205 Assig 2 Graphs Report

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Introduction and Background:

Within the academic field, one of the most important methods for validating theories is comparing multiple papers. When comparing many papers within those papers, there are often other papers that are referenced, which are important. For the chosen paper to be valid, often the papers referenced in it have to be somewhat valid also. There are a few problems with this currently. Firstly, a referenced paper could be from 100 years ago, when the theories were completely different from today

Another problem with this current model is that it is incredibly hard to remember a network of references. Even remembering how one paper is connected to another can be difficult when managing actual content as well

Methodology:

The citation network is modelled as a directed weighted graph using an adjacency list. Each paper is represented by a node, and each citation is an edge weighted by the difference in year in the referenced paper compared to the written paper. Adjacency lists were chosen over a matrix because citation graphs are sparse; most papers cite only a small number of others. making a matrix wasteful at $O(V^2)$ memory.

Within the testing, each paper is given an ID and a random year. The citations are then generated by having each paper select a random number or random papers while ensuring that all cited papers are in the past.

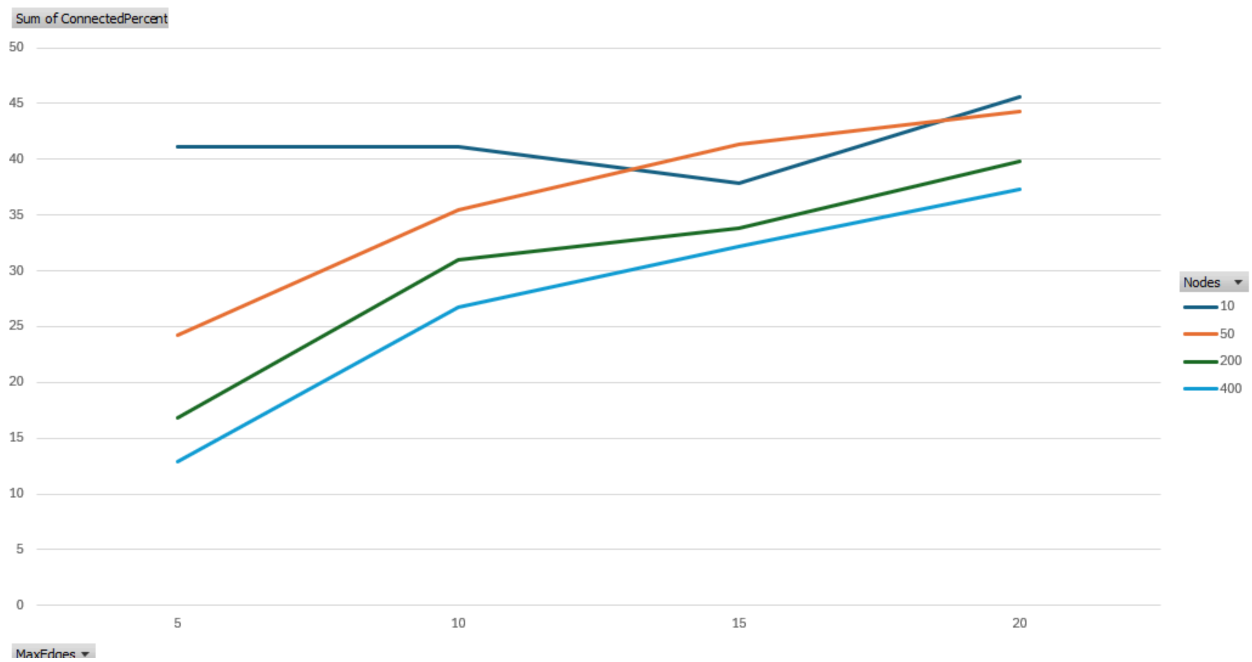
To implement a shortest path algorithm, I chose to use a Dijkstra's algorithm implementation. The pseudocode for Dijkstra's algorithm is as follows:

```
distance[all] = ∞, distance[src] = 0
for i = 0 to V-1:
        current = unvisited node with the smallest distance
        if distance[current] == ∞: break
        mark current visited
        for each edge (current → v, weight w):
                if distance[current] + w < distance[v]:
                        distance[v] = distance[current] + w
return distance[]
```

To show how edge density and graph size affect the connectivity and average path length Papers with counts of 100, 500, and 1000 were tested against maximum citation counts of 5, 10 and 15, with a fresh dataset generated for each combination.

Results and Discussion:

Here is a sample result from the testing



Visualisation of connected percentage vs number of citations for each number of papers test

=== INVESTIGATION: Graph Size vs Edge Count vs Avg Path Length ===

Nodes	MaxEdges	AvgPath	Connected%
10	5	25.32	41.1 %
10	10	21.11	41.1 %
10	15	20.65	37.8 %
10	20	23.26	45.6 %
50	5	22.04	24.2 %
50	10	22.62	35.4 %
50	15	22.26	41.3 %
50	20	20.39	44.3 %
200	5	25.16	16.8 %
200	10	23.43	31.0 %
200	15	20.17	33.8 %
200	20	19.32	39.8 %
400	5	27.27	12.9 %
400	10	20.33	26.7 %
400	15	21.63	32.2 %

400	20	20.94	37.3	%

```
=== Evaluation: Large Sized Citation Network ===

Graph parameters:
  Papers:  5000
  Authors: 5000
  Max citations per paper: 15

--- Graph Statistics ---
  Total edges:       39688
  Avg edges/paper:   7.94
  Max in-degree:     53
  Max out-degree:    15
  Isolated nodes:    0

--- Top 5 Most Cited Papers ---
  [1712] PaperID: 32873 | AuthorID: 424 | Year: 1970 | Citations: 53
  [2564] PaperID: 37928 | AuthorID: 17929 | Year: 1970 | Citations: 52
  [128] PaperID: 40929 | AuthorID: 10083 | Year: 1970 | Citations: 51
  [2274] PaperID: 38606 | AuthorID: 30317 | Year: 1970 | Citations: 51
  [460] PaperID: 40145 | AuthorID: 24262 | Year: 1970 | Citations: 48

--- Shortest Path Examples (Dijkstra's) ---
  From [769] Year:2008  To [3854] Year:1972  -->  Cost: 36
  From [1404] Year:2017  To [1678] Year:1994  -->  Cost: 23
  From [3015] Year:2013  To [1418] Year:1972  -->  Cost: 41
  From [2581] Year:2005  To [242] Year:1971  -->  Cost: 34

--- Connectivity Analysis ---
  Estimated connectivity: 24.6%

--- Average Shortest Path ---
  Average year-weighted path length: 22.50
```

The investigation tested node counts of 10, 50, 200, and 400 against maximum citation counts of 5, 10, 15, and 20, measuring average shortest path length and connectivity percentage for each combination.

Connectivity consistently increased with higher edge counts across all graph sizes. At 400 nodes, connectivity rose from 12.9% at maxEdges=5 to 37.3% at maxEdges=20, confirming that denser citation networks are more traversable. This pattern held regardless of graph size, suggesting edge density is the dominant factor in connectivity.

At 5000 nodes with maxEdges=15, the graph produced 39,688 edges, averaging 7.94 citations per paper, with a maximum in-degree of 53 and no isolated nodes. Estimated connectivity was 24.6% with an average path length of 22.50 years, consistent with the smaller graph trends.

It is important to note that the average citations were often lower than they would be hypothetically. This is likely due to the fact that papers cannot cite papers from the future, leading to older papers having less citations. In real life, however, this may not be so because there is no specific “start date” for when academic papers are written. On the other hand it could also be true that modern papers cite more references due to modern papers being more complex or due to a higher number of total papers

As can be theorised, the older papers are more often cited than the newer papers. This brings us back to our problem statement, where the oldest papers are most likely to be cited; they may also be most likely to be inaccurate or outdated, leading to the linked papers being inaccurate.

Future Improvements

If this project were to continue, it could be modified to use real-world data, incorporate topics/fields, and expanded to create a network visualisation. If this project were expanded to use real-world data and variables such as topic, it would open the door to calculating how networks of citations connect different topics and fields together while relating to authors and time periods. This would be very interesting to see visualised and could bring a new understanding and appreciation towards the fields

Conclusion:

This project successfully demonstrated how academic citation networks can be represented and analysed using graph data structures and algorithms. By modelling papers as nodes and citations as directed weighted edges, it was possible to investigate how papers are connected and how influence propagates through a research field over time.

The investigation showed that increasing the number of citations per paper has a significant impact on connectivity. Across all graph sizes tested, higher maximum citation counts resulted in a greater percentage of connected papers. While larger graphs tended to have lower overall connectivity, the same trend remained consistent as citation counts increased.

A notable observation was that the most highly cited papers were among the oldest papers in the network. This occurs because older papers have more opportunities to be cited by future work. Relating this to the original problem statement, it suggests that foundational papers can have a substantial influence on modern research long after publication. While many of these papers remain important, some may contain outdated assumptions or theories, meaning that inaccuracies can potentially propagate through chains of citations.

Overall, the project demonstrates that graph-based analysis provides a useful way of understanding citation relationships and identifying influential papers within a body of research. The results show how network structure changes as citation density increases and highlight the importance of considering both connectivity and publication age when evaluating academic literature. Future work could build upon this foundation by incorporating real citation datasets and visualising the resulting networks to better understand how ideas spread between authors, topics, and research fields.

References

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